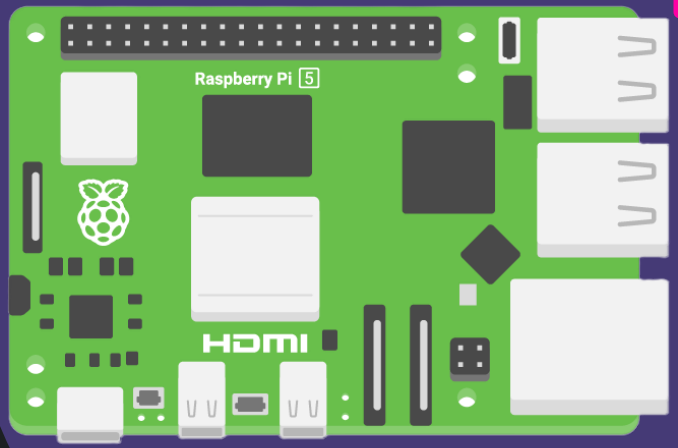
Retro Tv

Project



By: Haidar Ali Alshowaikh
Class: 8E

Matirials Used:

1- Rassperi Pi
2-Arduino Uno
3-Potentiometer 10K (A Knob)
4-Jumper Wires (Male to Female, and Female to Female)
5-Button
6-Joystick Module
7-A old Laptop screen
8- hdmi module

01

The Idea

How I got the idea

02

The Problems

The problems I faced during this project

03

Solution of Problems

How I solved these problems

04

The Wirring

How I wired the parts to the board

05

Goals

What are my project's goals.

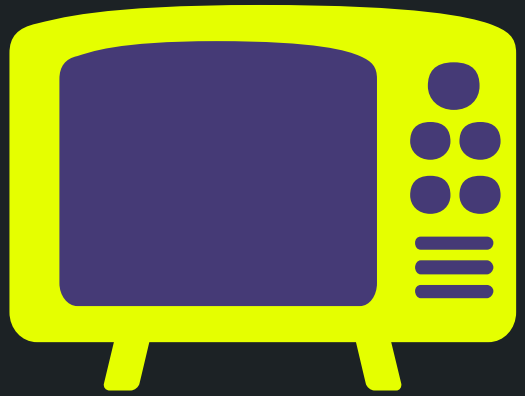
01

The Idea

How I got the idea



The Laptop...



I found a laptop that we don't use any more, So I thought of recycling it and giving it a new prepouse



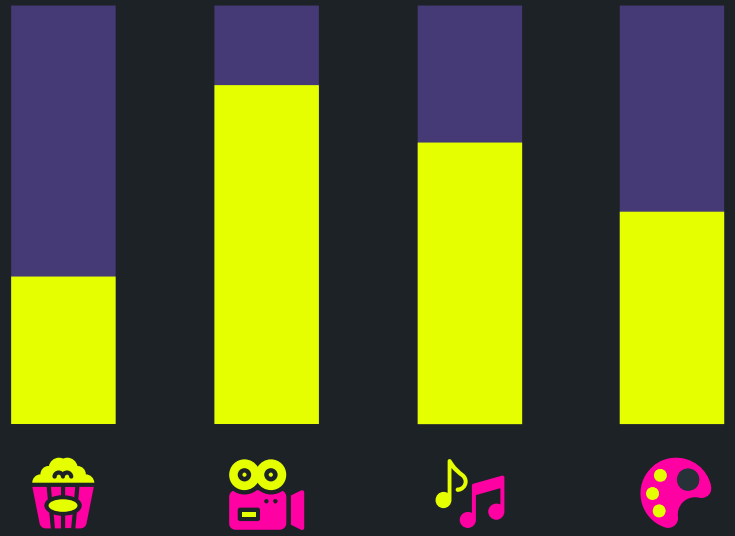
02

The Problems

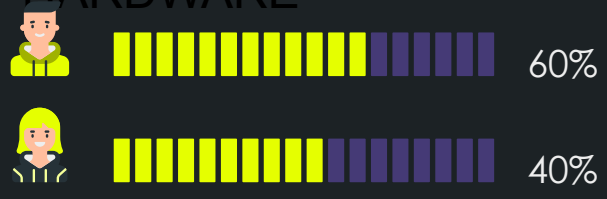
The problems I faced
during this project

PROBLEMS BY TYPE

CAUSES



SOFTWARE VS HARDWARE



TIME TO FIX



BIGGEST TIME SINKS

35%



Tracking down the PipeWire bug that sent audio out the wrong HDMI port

40%



Getting the joystick reliably read as arrow keys under the Wayland compositor

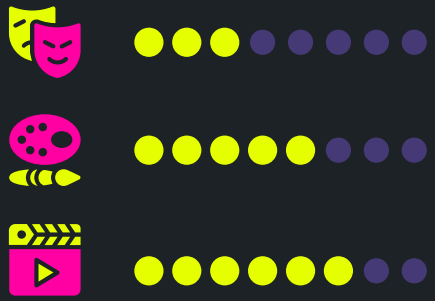
25%



Repairing the LCD panel after its LVDS connector latch broke

COMPONENT ISSUES

WHAT WENT WRONG





COMPONENT	ISSUE
Audio output	Wider than screws
Volume knob	Size unconfirmed
LCD panel	Grabbed by OS

COMPONENT	ISSUE
Joystick hole	Wrong HDMI port
Speaker grille	Both at once
Volume keys	Latch broke

FOUR MAIN PROBLEMS



AUDIO ROUTING

mpv was sending sound out the wrong HDMI port

VOLUME CONTROL

The knob changed app and system volume together





SCREEN CONNECTOR

The connector latch snapped during repair

HOLE VS SCREWS

Joystick hole ended up wider than its screws



03

Solution of Problems

How I solved these problems



HOW I FIXED THEM



VOLUME

Volume knob now calls wpctl to change the speaker

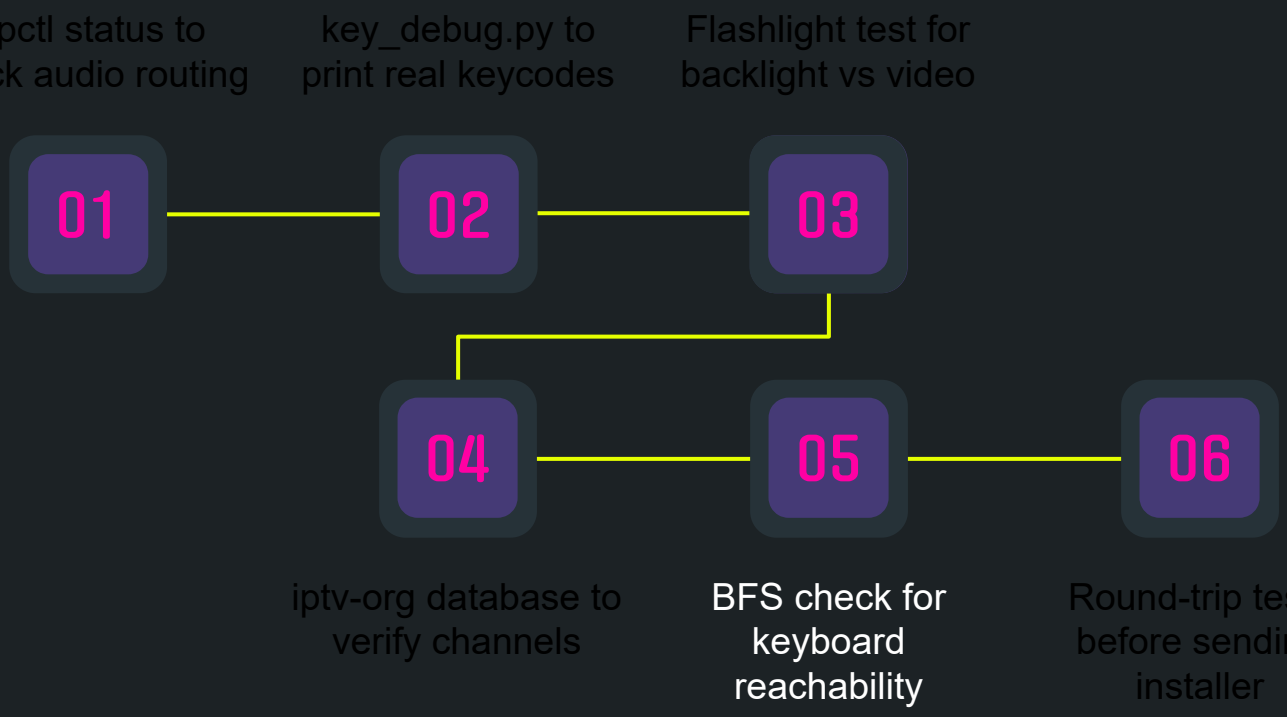
AUDIO DEVICE

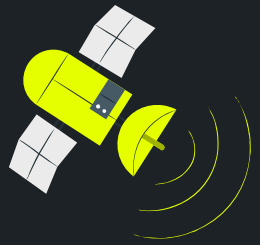
mpv now points straight at the correct ALSA device

KEY INPUT

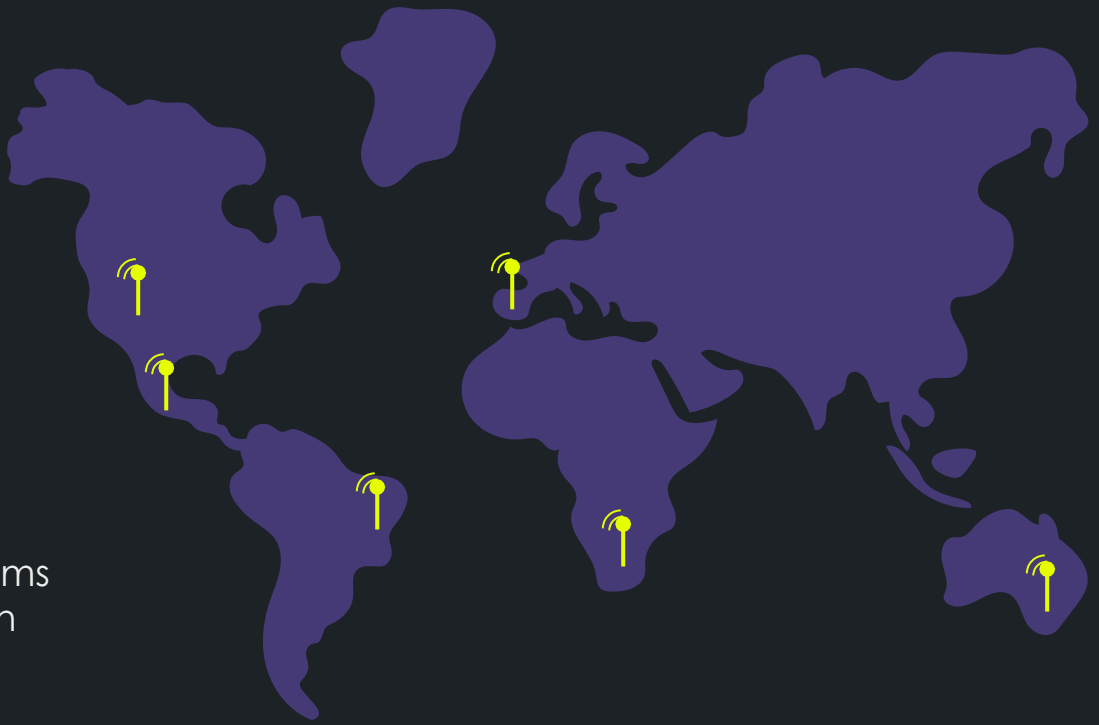
Checks the keysym and the raw hardware keycode both

DEBUGGING TOOLBOX





WHERE THIS STANDS NOW



Software problems are fixed. Screen repair is in progress.



04

The Wirring

How I wired the parts to the board

WIRING OVERVIEW

JOYSTICK

Wired to Arduino digital pins for direction, plus one for the click button

VOLUME KNOB

A 10K potentiometer wired to an Arduino analog input

POWER BUTTON

Its own Arduino digital pin, held for 2 seconds to shut down

ARDUINO -> PI

Connected over USB, sending plain text commands over serial

PI -> DRIVER BOARD

HDMI out from the Pi into the universal LCD controller board

DRIVER BOARD -> SCREEN

LVDS ribbon cable plus a separate backlight power connector

HOW A BUTTON PRESS TRAVELS



1. ARDUINO

Reads the joystick, knob, and button, and sends a plain text command over serial



2. THE BRIDGE

rpi_bridge.py on the Pi turns that command into a real key or mouse event



3. THE APP

tv.py receives it like any keyboard input and reacts on screen



WIRING IN PROGRESS

Every input component runs back to the Arduino on its own pin, and the Arduino talks to the Pi over a single USB serial cable - no extra wiring needed between the controller and the software.

FINISHED BUILD



THREE MAIN CONNECTIONS

USB SERIAL

Arduino to Raspberry Pi, carrying every joystick, knob, and button command

GPIO

The power button wired straight to the Pi for a direct, reliable shutdown

HDMI

Raspberry Pi to the driver board, carrying video to the recycled laptop screen



WIRING REFERENCE TABLE

ARDUINO	Joystick X/Y/click	Knob wiper	Button pin
PI	USB from Arduino	HDMI to driver board	GPIO to power button
DRIVER BOARD	HDMI in	LVDS out to panel	Backlight power out
SCREEN	LVDS in	Backlight in	-
POWER	5V to Pi	12V to driver board	Shared ground

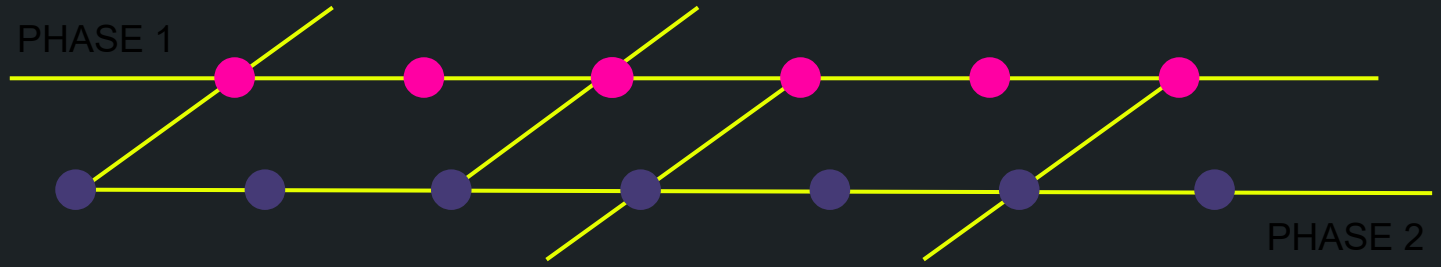
BUILD MILESTONES

Step 1

Wire inputs to the Arduino

Step 2

Flash sketch, test serial output



Step 3

Bridge turns serial into key events

Step 4

Wire board and screen into case

05

Goals

What are my project's goals.



GOALS



GOAL 1

Build a fully working retro TV controllable end-to-end with just the joystick



GOAL 2

Fit everything into a custom laser-cut enclosure sized for the recycled screen

264

Channels curated for the custom IPTV channel list, each checked twice before being added

THANKS!

Do you have any questions?
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